Problem 7 | Problems T6andT7_Electric and Magnetic Properties.pdf #7 | Intrinsic semiconductor · Arrhenius

Conductivity of Intrinsic Germanium at 75 °C

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7. The electrical conductivity of intrinsic germanium at 25 ºC is 2.22 $\Omega^{-1}m^{-1}$. What will its conductivity be at 75 ºC? The band gap of germanium is 0.67 eV and k_B = $8.62 \cdot 10^{-5}$ eV/K

Official Solution: 14.4 Ω⁻¹m⁻¹

HYPOTHESES & PARAMETERS

Hypotheses

1. **Intrinsic law.** In an intrinsic semiconductor the carrier number, and with it the conductivity, follows $\sigma = \sigma_0 \exp(-E_g/2k_BT)$ with the absolute temperature [Session 13 Slide 29]. The constant σ_0 is the same at both temperatures.
2. **Slide approximation.** The slide writes the proportionality $n_i \propto \exp(-E_g/2kT)$ and neglects the slower dependence of the prefactor and of the mobilities on the temperature; the exponential, which varies by orders of magnitude, dominates the ratio.
3. **Constant gap.** $E_g = 0.67$ eV is the same at 25 °C and 75 °C.
4. **Kelvin.** $T[\mathrm{K}] = T[^\circ\mathrm{C}] + 273.15$.

Data

Quantity	Symbol	Value as printed	In SI
Conductivity at 25 °C	σ_1	2.22 Ω⁻¹m⁻¹	2.22 Ω⁻¹m⁻¹
Temperatures	T_1, T_2	25 °C, 75 °C	298.15 K, 348.15 K
Band gap	E_g	0.67 eV	$0.67 \times 1.602 \times 10^{-19}$ J
Boltzmann constant	k_B	8.62×10^{-5} eV/K	$8.62 \times 10^{-5} \times 1.602 \times 10^{-19}$ J/K

Unknown: the conductivity σ_2 at 75 °C.

MODEL & STRATEGY

The unknown constant σ_0 is eliminated by taking the ratio of the law at the two temperatures; what remains is the exponential of $E_g/2k_B$ times the difference of the reciprocal temperatures. The straight line of $\ln\sigma$ against $1/T$ has the slope $-E_g/2k_B$, so the ratio is also the vertical distance between the two points of the figure.

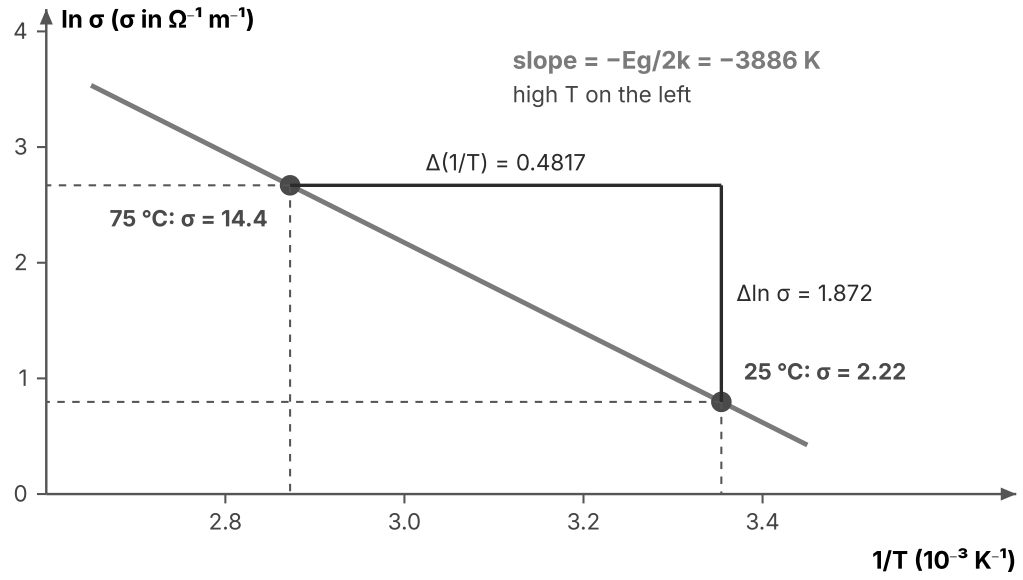


Figure 7. Problem 7: ln σ = ln σ0 − Eg/(2kT) for germanium (Eg = 0.67 eV); the points are the given 25 °C value and the computed 75 °C value, σ = 14.4 Ω⁻¹m⁻¹.

Strategy.

1. Write the law at T_1 and T_2 and divide.
2. Evaluate $E_g/2k_B$ in kelvin, the two reciprocals, their difference and the exponent.
3. Take the exponential and multiply by σ_1.

STEP-BY-STEP CALCULATION

1. The law at both temperatures [Session 13 Slide 29]

$$\sigma_1 = \sigma_0 \exp\left(-\frac{E_g}{2k_B T_1}\right) \qquad \sigma_2 = \sigma_0 \exp\left(-\frac{E_g}{2k_B T_2}\right)$$

2. Elimination of σ_0

Division of the second by the first and the rule $e^a/e^b = e^{a-b}$:

$$\frac{\sigma_2}{\sigma_1} = \frac{\exp\left(-\frac{E_g}{2k_B T_2}\right)}{\exp\left(-\frac{E_g}{2k_B T_1}\right)} = \exp\left[-\frac{E_g}{2k_B T_2} + \frac{E_g}{2k_B T_1}\right]$$

Common factor $E_g/2k_B$:

$$\sigma_2 = \sigma_1 \exp\left[\frac{E_g}{2k_B}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)\right]$$

3. Temperatures in kelvin

$$T_1 = 25 + 273.15 = 298.15\ \text{K} \qquad T_2 = 75 + 273.15 = 348.15\ \text{K}$$

4. The constant $E_g/2k_B$

Direct, with both quantities in eV (the eV cancels):

$$\frac{E_g}{2k_B} = \frac{0.67\ \text{eV}}{2 \times 8.62 \times 10^{-5}\ \text{eV/K}} = \frac{0.67}{1.724 \times 10^{-4}}\ \text{K} = 3886.3\ \text{K}$$

The same result in joules, as a check of the unit conversion ($1\ \text{eV} = 1.602 \times 10^{-19}$ J):

$$E_g = 0.67\ \text{eV} \times 1.602 \times 10^{-19}\ \frac{\text{J}}{\text{eV}} = 1.0733 \times 10^{-19}\ \text{J} \qquad k_B = 8.62 \times 10^{-5}\ \frac{\text{eV}}{\text{K}} \times 1.602 \times 10^{-19}\ \frac{\text{J}}{\text{eV}} = 1.3809 \times 10^{-23}\ \frac{\text{J}}{\text{K}}$$

$$\frac{E_g}{2k_B} = \frac{1.0733 \times 10^{-19}\ \text{J}}{2 \times 1.3809 \times 10^{-23}\ \text{J/K}} = 3886.3\ \text{K}$$

5. Reciprocal temperatures and their difference

$$\frac{1}{T_1} = \frac{1}{298.15\ \text{K}} = 3.35402 \times 10^{-3}\ \text{K}^{-1} \qquad \frac{1}{T_2} = \frac{1}{348.15\ \text{K}} = 2.87233 \times 10^{-3}\ \text{K}^{-1}$$

$$\frac{1}{T_1} - \frac{1}{T_2} = 3.35402 \times 10^{-3}\ \text{K}^{-1} - 2.87233 \times 10^{-3}\ \text{K}^{-1} = 4.8169 \times 10^{-4}\ \text{K}^{-1}$$

6. Exponent, exponential and result

$$\frac{E_g}{2k_B}\left(\frac{1}{T_1}-\frac{1}{T_2}\right) = 3886.3\ \mathrm{K}\times 4.8169\times 10^{-4}\ \mathrm{K}^{-1} = 1.8720$$

$$\frac{\sigma_2}{\sigma_1} = e^{1.8720} = 6.501$$

$$\sigma_2 = 2.22\ \Omega^{-1}\mathrm{m}^{-1}\times 6.501 = \mathbf{14.43\ \Omega^{-1}m^{-1}}$$

7. Using 273 instead of 273.15

$$\frac{1}{298}-\frac{1}{348} = 4.8214\times 10^{-4}\ \mathrm{K}^{-1} \qquad \sigma_2 = 2.22\times\exp(3886.3\times 4.8214\times 10^{-4}) = 2.22\times e^{1.8737} = 14.46\ \Omega^{-1}\mathrm{m}^{-1}$$

RESULT & CHECKS

Checks

- **Dimensionless exponent.**

$$\left[\frac{E_g}{2k_B}\left(\frac{1}{T_1}-\frac{1}{T_2}\right)\right] = \mathrm{K}\cdot\mathrm{K}^{-1} = 1$$

- **Slope.** On the plot of $\ln\sigma$ against $1/T$ the line has the slope -3886.3 K, and the vertical distance between the two points is $\ln(\sigma_2/\sigma_1) = 1.8720$.

$$\ln\frac{\sigma_2}{\sigma_1} = \ln\frac{14.43}{2.22} = \ln 6.501 = 1.8720$$

- **Consistency with the slide data.** Slide 31 gives $n_i = 2.4 \times 10^{19}$ m⁻³, $\mu_e = 0.39$ and $\mu_h = 0.19$ m²/(V·s) for germanium at room temperature:

$$\sigma_1 = n_i \, |q_e| \, (\mu_e + \mu_h) = 2.4 \times 10^{19} \times 1.602 \times 10^{-19} \times 0.58 \; \Omega^{-1}\mathrm{m}^{-1} = 2.23 \; \Omega^{-1}\mathrm{m}^{-1}$$

which agrees with the 2.22 of the statement.

Interpretation

A rise of 50 K multiplies the conductivity of germanium by 6.5: in a semiconductor heat creates carriers. At the limit $T_2 \to T_1$ the ratio tends to 1; at the limit $T_2 \to \infty$ the ratio saturates at the value of the next block (within the model of constant prefactor):

$$\frac{E_g}{2k_B T_1} = \frac{3886.3 \text{ K}}{298.15 \text{ K}} = 13.035 \qquad e^{13.035} = 4.58 \times 10^5$$

Check against the official solution

Item	Official	Computed	Comment
Conductivity at 75 °C	14.4 Ω⁻¹m⁻¹	14.43 Ω⁻¹m⁻¹	agrees with 273.15 K; with 273 K the result is 14.46, which rounds to 14.5, so the key uses 273.15